E.2 Paper Questions [142 marks]

1. [Maximum mark: 1]

A photoelectron is emitted when a photon of energy E is incident on a metal surface.

The work function of the surface is Φ.

What is the maximum possible speed of the photoelectron as it leaves the surface?

A. $\sqrt{\frac{2(E-\Phi)}{m_e}}$

B. $\sqrt{\frac{E-\Phi}{m_e}}$

C. $\sqrt{\frac{E-\Phi}{m_e h}}$

D. $\sqrt{\frac{2(E-\Phi)}{m_e h}}$ [1]

Markscheme
A

2. [Maximum mark: 1]

In a Compton scattering experiment a photon of wavelength λ_0 and frequency f_0 interacts with an electron.

After the interaction the wavelength and frequency of the photon are λ and f.

What are $\frac{\lambda}{\lambda_0}$ and $\frac{\lambda \times f}{\lambda_0 \times f_0}$?

	$\frac{\lambda}{\lambda_0}$	$\frac{\lambda \times f}{\lambda_0 \times f_0}$
A.	Greater than 1	Equal to 1
B.	Greater than 1	Less than 1
C.	Less than 1	Equal to 1
D.	Less than 1	Less than 1

[1]

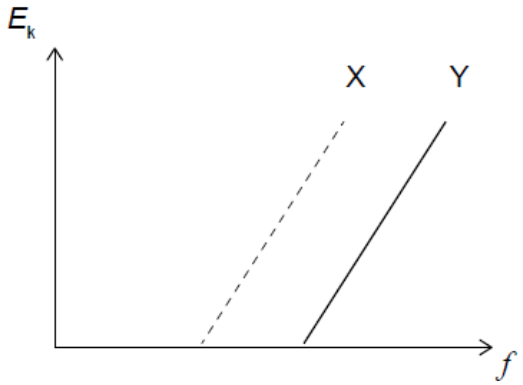
Markscheme

A

3. [Maximum mark: 1]

In two experiments X and Y measurements were made of the kinetic energies E_k of photoelectrons emitted when light is incident on two different metal surfaces.

The graph shows the variation of E_k with frequency f of the incident photons for both surfaces.



It can be deduced that in experiment Y…

A. the surface has a smaller work function than the surface in experiment X.

B. the surface has a larger work function than the surface in experiment X.

C. the intensity of the incident light is less than in experiment X. [1]

D. the intensity of the incident light is more than in experiment X.

Markscheme
B

4. [Maximum mark: 1]

The intensity of electromagnetic radiation incident on a surface is I. The wavelength of the radiation is doubled and the number of photons incident per unit time per unit area is doubled.

What is the new intensity of radiation incident on the surface?

A. $\frac{I}{2}$

B. I

C. $2I$

D. $4I$ [1]

Markscheme
B

5. [Maximum mark: 1]

Which of the following expressions gives the de Broglie wavelength, in metres, of a tennis ball of mass 0.06 kg and kinetic energy 3 J?

A. $\frac{h}{3}$

B. $\frac{h}{0.6}$

C. $\frac{h}{(0.6)^2}$

D. $\frac{h}{0.06}$ [1]

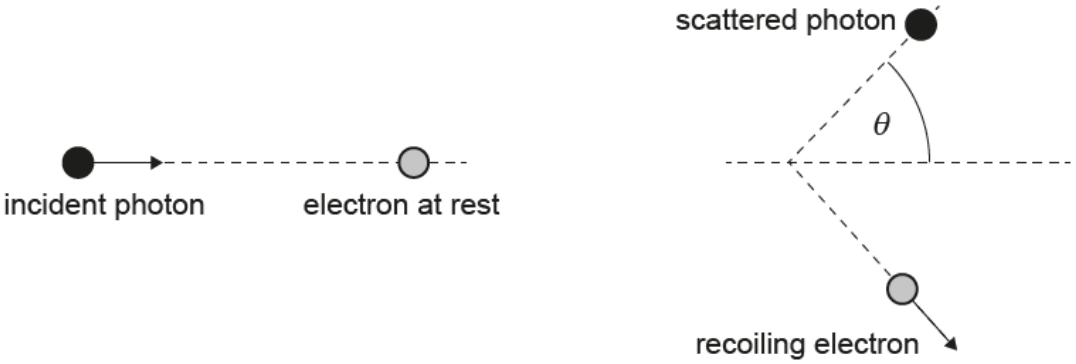
Markscheme
B

6. [Maximum mark: 8]

(a) Outline why Compton scattering offers better evidence for the particle nature of light than the photoelectric effect. [2]

Markscheme
Compton scattering provides evidence for «the energy and» the momentum of the photon / refers to the collision between photon and electron ✓ The photelectric effect provides evidence for the energy of photons ✓

A photon of wavelength 6.40×10^{-12} m and energy 0.194 MeV is incident on an electron at rest. The photon scatters at an angle θ with wavelength 7.47×10^{-12} m. The electron recoils.



(b) Outline why the wavelength of the scattered photon is longer than that of the incident photon. [2]

Markscheme

The photon loses energy / transfers energy to the electron ✓

$E = \frac{hc}{\lambda}$ / lower energy implies longer wavelength ✓

(c.i) Show that the energy of the scattered photon is about 0.17 MeV. [1]

Markscheme

$E = \text{«}\frac{hc}{\lambda} \text{=»} \frac{1.24\times10^{-6}}{7.47\times10^{-12}}$ **OR** $0.166\,MeV$ ✓

(c.ii) Calculate the kinetic energy of the recoiling electron. [1]

Markscheme

$E = \text{«}0.194 - 0.166 \text{=»} 0.028\,MeV$ ✓

Accept 0.024 from the value given as show that.

Allow ECF from c) i)

Accept 4.5 x 10^{-15} J.

(c.iii) For an electron, $\frac{h}{m_e c} = 2.43 \times 10^{-12}\,\text{m}$ Determine θ. [2]

Markscheme

$7.47 \times 10^{-12} - 6.40 \times 10^{-12} = 2.43 \times 10^{-12}(1 - \cos\theta)$ ✓

«$\cos\theta = 0.5597$ so» $\theta = 56.0^{\circ}$ ✓

Accept angle in radians (0.98 « rad »).

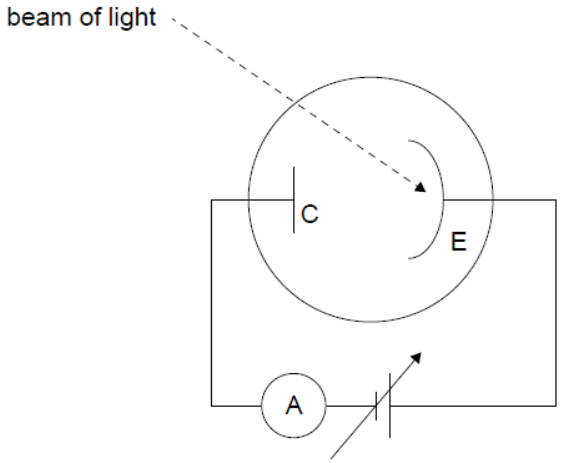
Award [2] if 56 «°» or 0.98 « rad » are seen as the answer without working

7. [Maximum mark: 6]

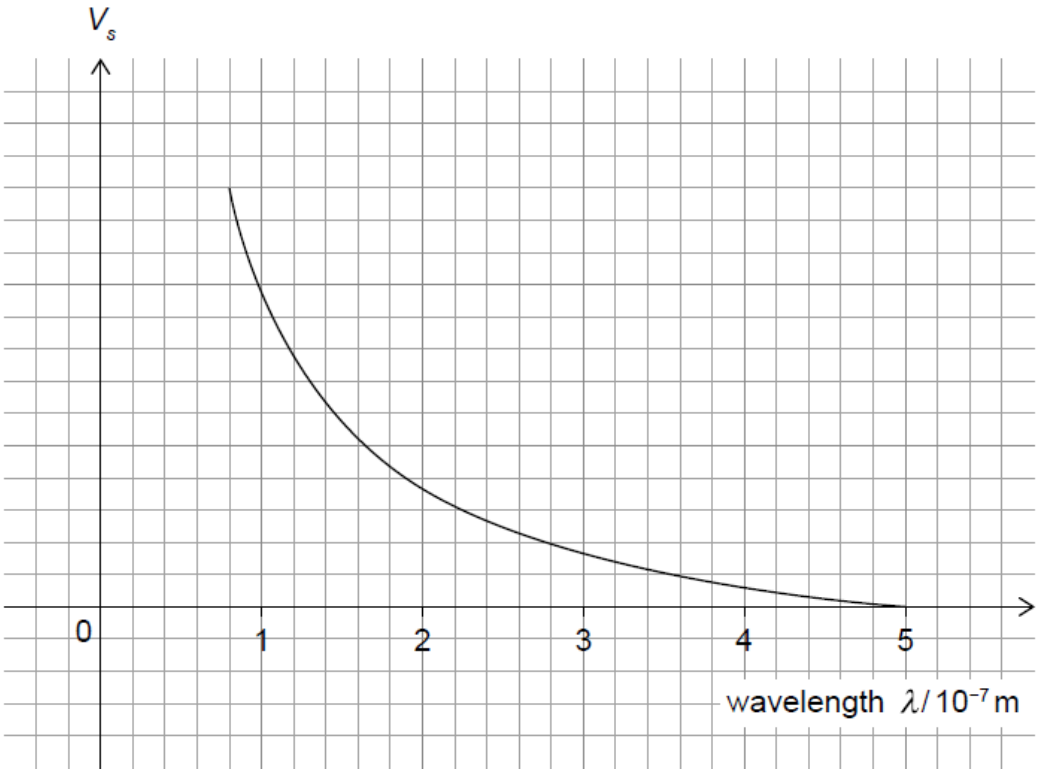
An electric circuit is used to investigate the photoelectric effect.

A beam of light from a laser of variable wavelength λ and constant intensity I is used to illuminate a curved metallic electrode E. Another electrode, C, is near E. Both electrodes are in vacuum. For each λ tuned in the laser, the voltage across CE is raised. The minimum voltage V_s that shows no current in the ammeter is recorded.

diagram not to scale



The graph shows the variation of V_s with λ.



(a) Show that the work function of the metal used in C is about 2.5 eV. [2]

Markscheme

reads λ= 5.0x10^{-7} «m» ✓

ϕ =«h c /λ =» $\frac{1.24\times10^{-6}}{5\times10^{-7}}$ ***OR*** 2.48 «eV» ✓

Must see full substitution or answer to 3 or more significant figures for MP2.

(b) Calculate V_s when λ is 4.0 × 10^{-7} m. [2]

Markscheme

ALT 1

«hf ***OR*** hc/λ= (1.24 × 10^{-6})/(4.0 × 10^{-7}) = » 3.1 «V» ✓

«3.1 – 2.5 =» 0.6 «V» ✓

ALT 2

$\frac{(6.63\times10^{-34})(3\times10^{8})}{(4\times10^{-7})} - (4 \times 10^{-19}) = 9.73 \times 10^{-20}$ «J» ✓

«(9.73 × 10^{-20})/(1.6 × 10^{-19})» = 0.6 «V» ✓

The laser is now replaced by a monochromatic laser of red light and variable intensity.

Although the intensity of the laser is increased significantly, no current is read in the ammeter.

(c) Explain how this observation is consistent with the particle nature of light. [2]

Markscheme

Particle nature suggests photons have energy related to frequency/wavelength ✓

Intensity does not change photon energy

OR

Intensity not related to E necessary to release electron. ✓

One photon of sufficient energy is necessary to release an electron

OR

photon must have sufficient energy to release electron. ✓

If light is a wave, higher intensity should provide energy with time to eject an electron. ✓

8. [Maximum mark: 1]

A photon of wavelength λ scatters off an electron at rest. The scattered photon has wavelength λ'.

What is the fraction of the incident photon energy that gets transferred to the electron?

A. $\frac{\lambda}{\lambda'}$

B. $\frac{\lambda'}{\lambda}$

C. $\frac{\lambda'-\lambda}{\lambda}$

D. $\frac{\lambda'-\lambda}{\lambda'}$ [1]

Markscheme

D

9. [Maximum mark: 1]

Monochromatic light of frequency f_1 is incident on the surface of a metal. The stopping voltage for this light is V_1. When the frequency of the radiation is changed to f_2, the stopping voltage is V_2.

What is the quantity $\frac{V_2 - V_1}{f_2 - f_1}$ equal to?

A. h

B. $\frac{h}{e}$

C. $\frac{h}{c}$

D. $\frac{hc}{e}$ [1]

Markscheme

B

10. [Maximum mark: 1]

Which statement is correct about Compton scattering of a photon by an electron?

A. The energy of the photon decreases.

B. The wavelength of the photon decreases.

C. The momentum of the photon is unchanged.

D. The combined momentum of the particles increases.

[1]

Markscheme
A

11. [Maximum mark: 1]

A proton and an alpha particle are accelerated by the same electric potential difference. The kinetic energy of the proton is E and its de Broglie wavelength is λ. What is the kinetic energy and the de Broglie wavelength of the alpha particle?

	Kinetic energy of alpha particle	**De Broglie wavelength of alpha particle**
A.	E	less than λ
B.	E	greater than λ
C.	greater than E	less than λ
D.	greater than E	greater than λ

[1]

Markscheme
C

12. [Maximum mark: 1]

An atom of hydrogen ($^1_1\mathrm{H}$) and an atom of helium ($^4_2\mathrm{He}$) are moving with the same kinetic energy.

The de Broglie wavelength of the hydrogen atom is λ_{H} and the de Broglie wavelength of the helium atom is λ_{He}.

What is $\frac{\lambda_{\mathrm{H}}}{\lambda_{\mathrm{He}}}$?

A. $\frac{1}{2}$

B. 1

C. 2

D. 4 [1]

Markscheme
C

13. [Maximum mark: 1]
An electron is accelerated from rest through a potential difference of 3.8 kV.

The de Broglie wavelength of the electron after acceleration is

A. 0.021 mm

B. 0.021 μm

C. 0.021 nm

D. 0.021 pm [1]

Markscheme
C

14. [Maximum mark: 1]
The Compton effect can be explained using

A. conservation of momentum

B. kinematic equations applied in two dimensions

C. the concept of a photon

D. the wave theory of light [1]

Markscheme
A

15. [Maximum mark: 1]
A beam of X-rays of wavelength 100.00 pm is scattered from a block of carbon. Radiation is observed at right angles to the incident beam.

What is the Compton shift for the observed radiation?

A. 0.1024 pm

B. 2.4322 pm

C. 2.4322 nm

D. 0.1024 nm [1]

Markscheme
B

16. [Maximum mark: 8]
A beam of electrons is incident on a thin crystalline sample of graphite. The electrons emerging from the sample form a pattern on a fluorescent screen. The

pattern consists of a series of bright and dark rings concentric with the direction of the incident beam.

(a) Outline why the pattern observed on the screen is an evidence for matter waves. [2]

Markscheme

The pattern is formed when the electrons scattered from adjacent planes in the graphite crystal undergo interference / diffract ✓

Interference / diffraction is a property of waves only ✓

The beam is produced by accelerating electrons through an electric potential difference *U*.

(b) A typical interatomic distance in the graphite crystal is of the order of 2×10^{-10} m. Estimate the minimum value of *U* for the pattern in (a) to be formed on the screen. [4]

Markscheme

The de Broglie wavelength of the electrons should be comparable to or shorter than the interatomic distance / 2×10^{-10} m ✓

Momentum of electrons $\approx \frac{6.63\times10^{-34}}{2\times10^{-10}}$ «$= 3.32 \times 10^{-2}$ N s» ✓

Kinetic energy of electrons = «$\frac{p^2}{2m} = \frac{(3.32\times10^{-24})^2}{2\times9.11\times10^{-31}}$ =»6.0×10^{-18} «J» ✓

U = «$\frac{E}{e} = \frac{6.10\times10^{-18}}{1.6\times10^{-19}}$ =»40 «V» ✓

(c) Protons can also be accelerated by the same potential difference *U*. Compare, without calculation, the de Broglie wavelength of the protons to that of the electrons. [2]

Markscheme

The protons have the same energy but greater mass hence a greater momentum than the electrons ✓

From $\lambda = \frac{h}{p}$, the protons will have a shorter wavelength ✓

17. [Maximum mark: 7]

In a Compton scattering experiment, an X-ray photon interacts with a free, stationary electron. The electron recoils with an energy of 500 eV. The wavelength of the scattered photon is 7.80×10^{-11} m.

(a) Show that the energy of the scattered photon is about 16 keV. [1]

Markscheme

$\frac{1.24 \times 10^{-6}}{7.80 \times 10^{-11}}$ «= 15.9 keV» ✓

(b) Determine the wavelength of the incident photon. [2]

Markscheme

Energy of incident photon =«$15.9 \times 10^3 + 500 =$»16.4×10^3 «eV» ✓

Wavelength of incident photon =«$\frac{1.24 \times 10^{-6}}{16.4 \times 10^3} =$»$7.56 \times 10^{-11}$ «m» ✓

(c) Outline why the results of the experiment are inconsistent with the wave model of electromagnetic radiation. [2]

Markscheme

The wavelength of the X-rays changes ✓

According to the wave model, the wavelength of the incident and scattered X-rays should be the same ✓

(d) Calculate the scattering angle of the photon. [2]

Markscheme

$(7.80 - 7.56) \times 10^{-11} = \frac{1.24 \times 10^{-6}}{0.511 \times 10^{6}}(1 - \cos\theta)$ ✓

$\theta = 89^{\circ}$ ✓

18. [Maximum mark: 8]

(a) The quantity $\frac{h}{m_e c}$ is known as the Compton wavelength.

Show that the Compton wavelength is about 2.4 pm. [1]

Markscheme

2.43 pm ✓

(b) A photon with a wavelength of 6.00 pm interacts with a stationary electron. After the interaction the photon's wavelength has changed by 2.43 pm.

(b.i) State the wavelength of the photon after the interaction. [1]

Markscheme

8.43 pm ✓

(b.ii) Outline why the wavelength of the photon has changed. [2]

Markscheme

(Energy of photon inversely prop to wavelength)

photon transfers some of its energy to the electron. ✓

If its energy decreases so its wavelength increases. ✓

(b.iii) Deduce the scattering angle for the photon. [2]

Markscheme

(for this interaction)

$\Delta\lambda = \frac{h}{m_e c}(1 - \cos\theta) = \frac{h}{m_e c}$ and therefore $(1 - \cos\theta)$ must equal 1 ✓

so cos theta = 0 and theta = 90 deg ✓

(b.iv) Determine, in J, the kinetic energy of the electron after the interaction. [2]

Markscheme

(Because energy is conserved)

$E_k = hc\left(\frac{1}{\lambda_i} - \frac{1}{\lambda_f}\right)$

$= 6.63 \times 10^{-34} \times 3.00 \times 10^8 \left(\frac{1}{6.00\times10^{-12}} - \frac{1}{8.43\times10^{-12}}\right)$ ✓

9.6 fJ ✓

19. [Maximum mark: 8]

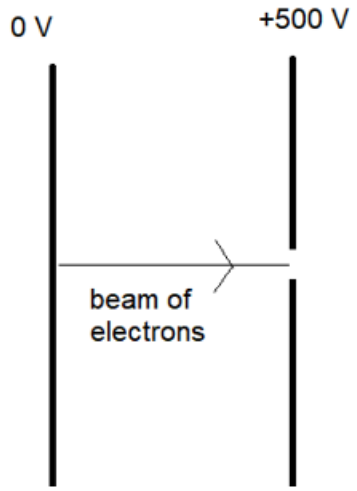
(a) State the de Broglie hypothesis. [2]

Markscheme

a moving particle has wave properties ✓

de Broglie wavelength = Planck constant÷momentum (must define p if quoted as equation) ✓

A beam of electrons is accelerated from rest through a potential difference of 500 V maintained between two plates in a vacuum. The electrons then travel through a circular hole in the +500 V plate.



(b) Calculate the maximum speed of the electrons in the beam. [2]

Markscheme

use of $\frac{1}{2} mv^2 = eV$ ✓

1.33×10^7 m s^{-1} ✓

(c) After passing through the circular hole the electrons strike a fluorescent screen.

Predict whether an apparatus such as this can demonstrate that moving electrons have wave properties. [4]

Markscheme

idea that de Broglie wavelength and hole size must be similar ✓

Use of $\lambda = \frac{h}{\sqrt{2meV}}$ ✓

Leading to λ around 5 x 10^{-11} m (which is unrealistic for a practical situation) ✓

It would not be possible to construct this hole
OR
hole must be smaller than an atom so impossible ✓

20. [Maximum mark: 1]
Electrons are accelerated through a potential difference and are incident on a sample of crystal atoms causing a diffraction pattern.

When the potential difference is increased, the maxima

A. are closer together.

B. are farther apart.

C. have a similar intensity.

D. are unchanged. [1]

Markscheme

A

21. [Maximum mark: 7]

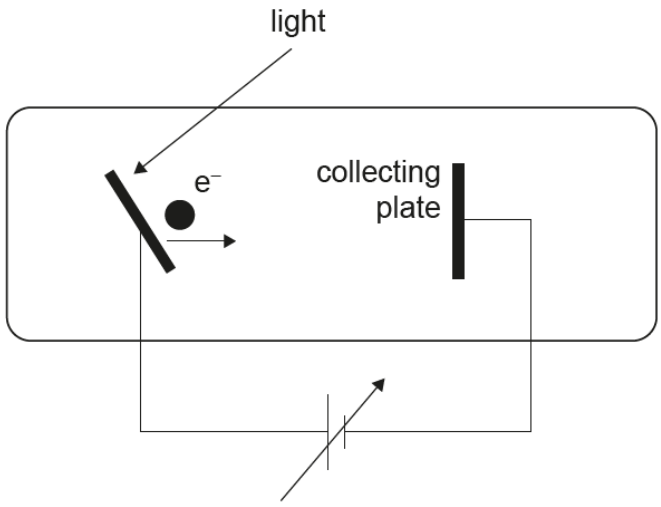
(a) Electrons are emitted from a metal surface when very low intensity light is incident on it. This emission occurs with no observable time delay.

Outline why this observation is not consistent with the wave theory of light. [2]

Markscheme

«in the wave theory of light » low intensity light means low «rate of» transfer of energy. ✔
and so electrons would need time to accumulate the energy they need to escape «leading to an observable time delay» ✔

(b) Light of wavelength 5.2×10^{-7} m is incident on a metal surface and causes the emission of electrons of maximum kinetic energy 1.8 eV.



(b.i) Estimate, in eV, the work function of the metal surface. [2]

Markscheme

$1.8 = \frac{1.24 \times 10^{-6}}{5.2 \times 10^{-7}} - \phi$ ✔

ϕ = 0.59 eV ✔

Accept 0.58 or 0.59.

(b.ii) The emf of the power supply is set at 0.60 V. Determine the speed of the electrons as they arrive at the collecting plate. [3]

Markscheme

KE of electrons at collecting plate is «1.8 – 0.6=» 1.2 eV ✔

$\frac{1}{2} \times 9.1 \times 10^{-31} \times v^2 = 1.2 \times 1.6 \times 10^{-19}$ ✔

v = 6.5 × 10^5 «m s^{-1}» ✔

Award [2] for $v = 9.2 \times 10^5$ m s^{-1} (using KE = 2.4 eV).

22. [Maximum mark: 1]

Light of intensity *I* and frequency f, which is above the threshold frequency, is directed at a polished metal surface. The following three claims are made about electrons ejected from the surface.

I. The ejection is almost instantaneous.

II. The number ejected is proportional to *I*.

III. Their kinetic energy is equal to Planck's constant multiplied by f.

Which statements are correct?

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

[1]

Markscheme
A

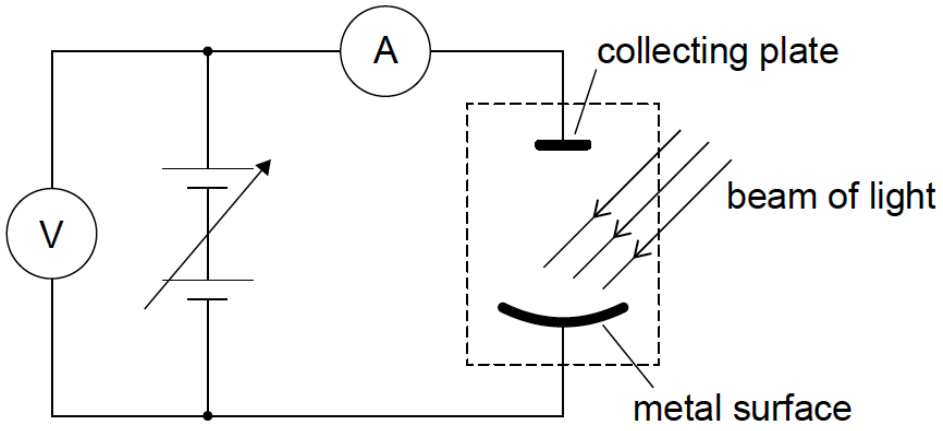
23. [Maximum mark: 9]

In an experiment to investigate the photoelectric effect, a beam of monochromatic light is incident on a metal surface. When the frequency of light is less than a certain threshold frequency f_0, no electrons are emitted from the surface.

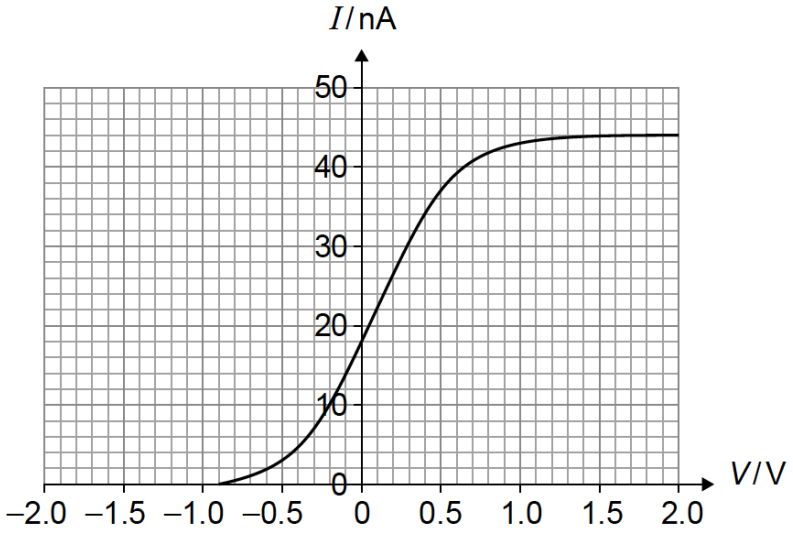
(a) Outline how this observation supports the photon model of light. [2]

Markscheme
photon energy is proportional to frequency ✔ «when $f < f_0$,» photon energy is less than the work function/energy needed to release an electron «from the surface» ✔ *Need to see 'is proportional to', not 'is related to'. Reference to E=hf is satisfactory for MP1*

The collecting plate of the photoelectric cell is at a variable potential V relative to the metal surface.



The graph shows the variation of the current *I* in the ammeter with *V*, for light of wavelength 430 nm.



(b) Explain why *I*

(b.i) is zero for large negative *V*. [2]

Markscheme

electrons are repelled from/do not « have enough KE to » reach the collecting plate /attracted back to the photosurface ✔

maximum KE of an electron is less than the work/energy needed to transfer it to the collecting plate ✔

Stopping potential /reverse pd opposes electron flow/reduces I « to zero » to the photocathode. ✔

(b.ii) approaches a constant value for large positive *V*. [2]

Markscheme

ALTERNATIVE 1

electrons are attracted to the collecting plate ✔

Every electron emitted from the surface reaches the collecting plate ✔

ALTERNATIVE 2

there is a maximum number of photoelectrons that can be emitted per second ✔

this number is limited by the intensity of incident radiation «not the applied voltage» ✔

Select responses from one ALT only

(c) Calculate, in eV

(c.i) the energy of a photon of wavelength 430 nm. [1]

Markscheme

hf =« $\frac{hc}{\lambda} = \frac{1.24 \times 10^{-6}}{430 \times 10^{-9}}$ =»2.9 «eV» ✔

(c.ii) the work function of the metal surface. [2]

Markscheme

E_K =0.9 «eV» ✔

Φ = «hf – E_k = 2.0 «eV» ✔

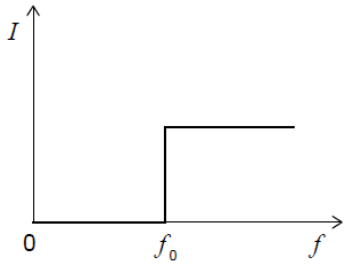
Look for 2.9–0.9 = 2.0 eV

Award [2] for BCA

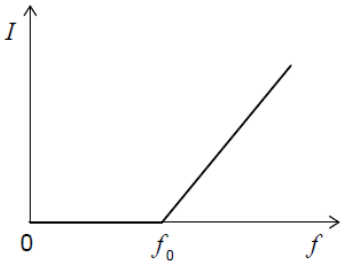
24. [Maximum mark: 1]

Light falls on a metallic surface whose threshold frequency is f_0. The number of photons incident per second is constant. Which graph correctly shows how the current I emitted from the surface depends on the frequency f of the incident light?

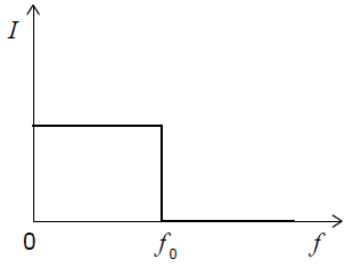
A.



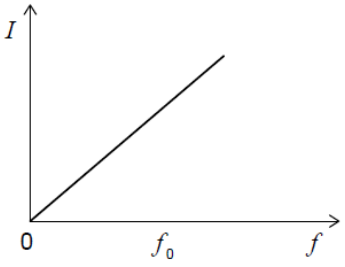
B.



C.



D.



[1]

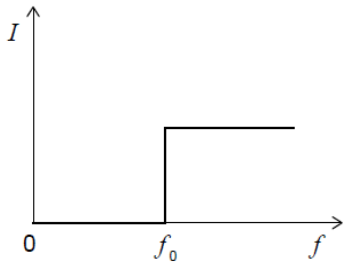
Markscheme

A

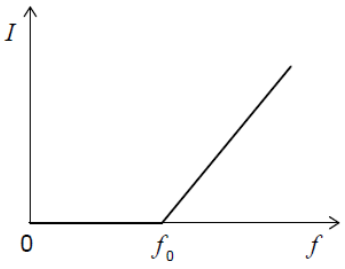
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Light falls on a metallic surface whose threshold frequency is f_0. The number of photons incident per second is constant. Which graph correctly shows how the current I emitted from the surface depends on the frequency f of the incident light?

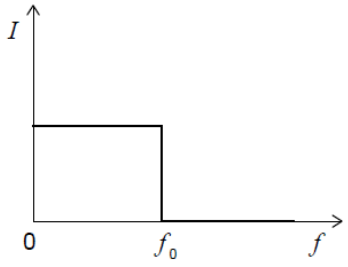
A.



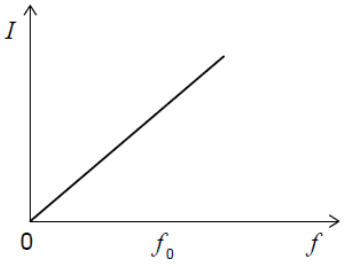
B.



C.



D.



[1]

Markscheme

A

26. [Maximum mark: 1]

Light of frequency f is incident on a metallic surface of work function W. Photoelectrons with a maximum kinetic energy E_{max} are emitted. The frequency of the incident light is changed to $2f$.

What is true about the maximum kinetic energy and the work function?

	Maximum kinetic energy	Work function
A.	less than $2E_{max}$	unchanged
B.	less than $2E_{max}$	greater than W
C.	greater than $2E_{max}$	unchanged
D.	greater than $2E_{max}$	greater than W

[1]

Markscheme

C

27. [Maximum mark: 8]

(a) Photons of wavelength 468 nm are incident on a metallic surface.
The maximum kinetic energy of the emitted electrons is 1.8 eV.

Calculate

(a.i) the work function of the surface, in eV. [2]

Markscheme

Use of $E_{max} = \frac{hc}{\lambda} - \phi \Rightarrow \phi = \frac{hc}{\lambda} - E_{max}$ ✓

$\phi = \text{«}\frac{hc}{\lambda} - E_{max} = \frac{(6.63\times10^{-34})(3\times10^{8})}{(468\times10^{-9})(1.6\times10^{-19})} - 1.81\text{»} = 0.85625 \approx 0.86$ «eV» ✓

(a.ii) the longest wavelength of a photon that will eject an electron from this surface. [2]

Markscheme

Use of $\frac{hc}{\lambda} = \phi \Rightarrow \lambda = \frac{hc}{\phi}$ ✓

$\lambda =$« $\frac{(6.63\times10^{-34})(3\times10^{8})}{(468\times10^{-9})(1.6\times10^{-19})}$ =» 1.45×10^{-6} «m» ✓

*Allow **ECF** from **a(i)***

(b.i) In an experiment, alpha particles of initial kinetic energy 5.9 MeV are directed at stationary nuclei of lead ($^{207}_{82}\mathrm{Pb}$). Show that the distance of closest approach is about 4×10^{-14} m. [2]

Markscheme

2e ***AND*** 82e seen

OR

3.2×10^{-19} «C» ***AND*** 1.312×10^{-17} «C» seen ✓

$d = \frac{8.99\times10^{9}\times(2e)(82e)}{5.9\times10^{6}\times e} = 3.998 \times 10^{-14} \approx 4 \times 10^{-14}$ «m» ✓

*Must see either clear substitutions or answer to at least 4 s.f. for **MP2**.*

(b.ii) The radius of a nucleus of $^{207}_{82}\mathrm{Pb}$ is 7.1×10^{-15} m. Explain why there will be no deviations from Rutherford scattering in the experiment in (b)(i). [2]

Markscheme

The closest approach is «significantly» larger than the radius of the nucleus / far away from the nucleus/***OWTTE***. ✓

«Therefore» the strong nuclear force will not act on the alpha particle.✓

28. [Maximum mark: 1]

Monochromatic electromagnetic radiation ejects photoelectrons from a metal surface. The minimum frequency for which this is possible is f.

When radiation of frequency $2f$ is incident on the surface, the kinetic energy of the photoelectrons is K.

What is the kinetic energy of the photoelectrons when the frequency of the radiation is $4f$?

A. K

B. $2K$

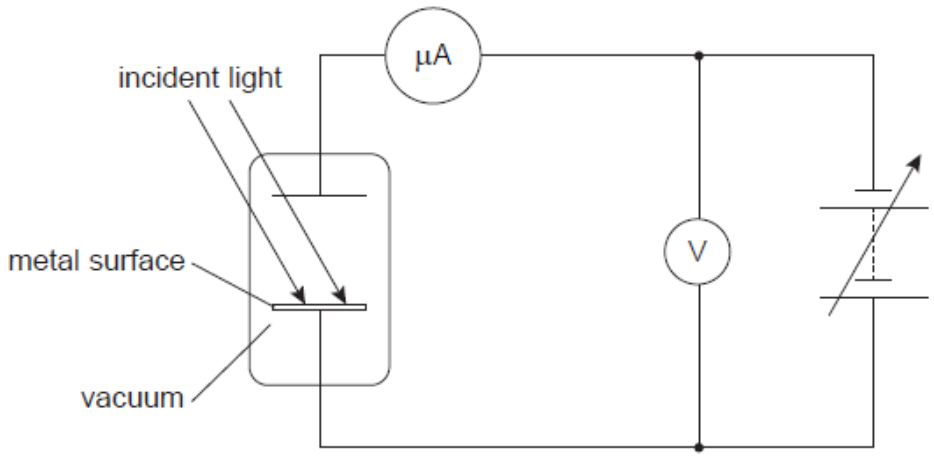
C. $3K$

D. $4K$ [1]

Markscheme
C

29. [Maximum mark: 1]

Light with photons of energy 8.0×10^{-20} J are incident on a metal surface in a photoelectric experiment.



The work function of the metal surface is 4.8×10^{-20} J . What minimum voltage is required for the ammeter reading to fall to zero?

A. 0.2 V

B. 0.3 V

C. 0.5 V

D. 0.8 V

[1]

Markscheme
A

30. [Maximum mark: 1]
What is evidence for wave–particle duality?

A. Line spectra of elements

B. Electron-diffraction experiments

C. Rutherford alpha-scattering experiments

D. Gamma-ray spectra

[1]

Markscheme
B

31. [Maximum mark: 1]
Three correct statements about the behaviour of electrons are:

I. An electron beam is used to investigate the structure of crystals.
II. An electron beam produces a pattern of fringes when sent

through two narrow parallel slits.
III. Electromagnetic radiation ejects electrons from the surface of a metal.

Which statements are explained using the wave-like properties of electrons?

A. I and II only

B. I and III only

C. II and III only

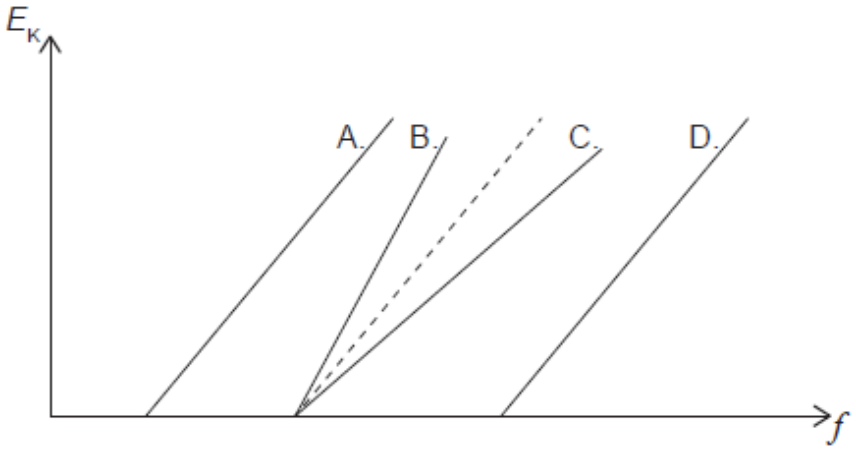
D. I, II and III

[1]

Markscheme
A

32. [Maximum mark: 1]
The dashed line represents the variation with incident electromagnetic frequency f of the kinetic energy E_K of the photoelectrons ejected from a metal surface. The metal surface is then replaced with one that requires less energy to remove an electron from the surface.

Which graph of the variation of E_K with f will be observed?



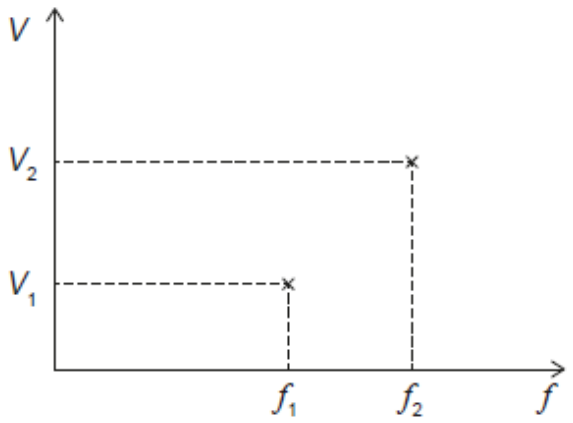
[1]

Markscheme

A

33. [Maximum mark: 1]

In a photoelectric experiment a stopping voltage V required to prevent photoelectrons from flowing across the photoelectric cell is measured for light of two frequencies f_1 and f_2. The results obtained are shown.



The ratio $\frac{V_2-V_1}{f_2-f_1}$ is an estimate of

A. e

B. h

C. $\frac{e}{h}$

D. $\frac{h}{e}$ [1]

Markscheme

D

34. [Maximum mark: 1]

An electron of non-relativistic speed v interacts with an atom. All the energy of the electron is transferred to an emitted photon of frequency f. An electron of speed $2v$ now interacts with the same atom and all its energy is transmitted to a second photon. What is the frequency of the second photon?

A. $\frac{f}{4}$

B. $\frac{f}{2}$

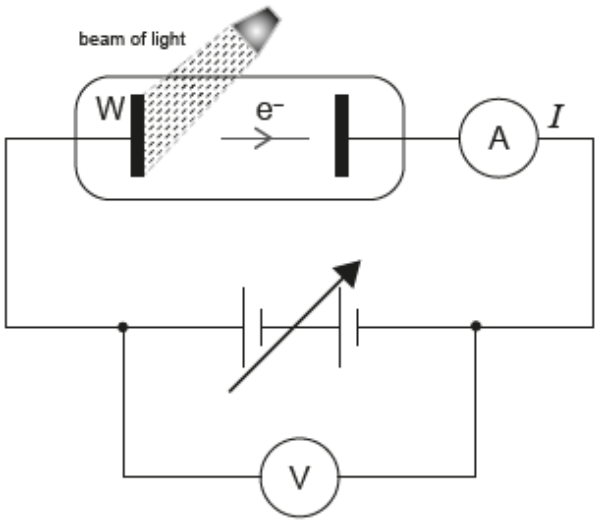
C. $2f$

D. $4f$ [1]

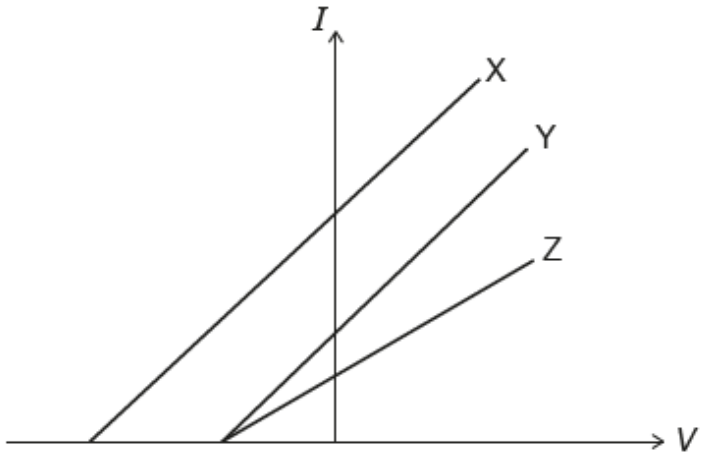
Markscheme
D

35. [Maximum mark: 1]

In a photoelectric effect experiment, a beam of light is incident on a metallic surface W in a vacuum.



The graph shows how the current I varies with the potential difference V when three different beams X, Y, and Z are incident on W at different times.



I. X and Y have the same frequency.
II. Y and Z have different intensity.
III. Y and Z have the same frequency.

Which statements are correct?

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

[1]

Markscheme
C

36. [Maximum mark: 1]
What is true for the Bohr model for the hydrogen atom?

A. Angular momentum of electrons is quantized.

B. Electrons are described by wave functions.

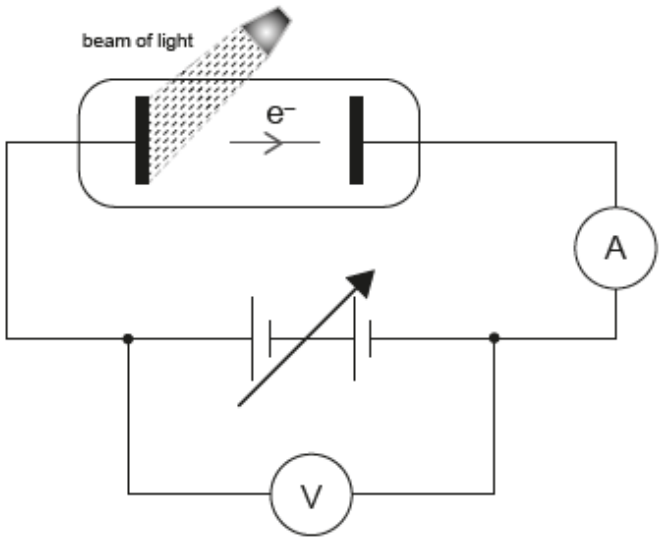
C. Electrons never exist in fixed orbitals.

D. Electrons will continuously emit radiation. [1]

Markscheme
A

37. [Maximum mark: 7]

In an electric circuit used to investigate the photoelectric effect, the voltage is varied until the reading in the ammeter is zero. The stopping voltage that produces this reading is 1.40 V.



(a) Describe the photoelectric effect. [2]

Markscheme
electrons are ejected from the surface of a metal ✓ after gaining energy from photons/electromagnetic radiation ✓

there is a minimum «threshold» energy/frequency
OR
maximum «threshold» wavelength ✓

(b) Show that the maximum velocity of the photoelectrons is $700\,\text{km s}^{-1}$. [2]

Markscheme

«$eV = \frac{1}{2}mv^2$» and manipulation to get v ✓

$v = \sqrt{\frac{2\times1.6\times10^{-19}\times1.4}{9.1\times10^{-31}}}$ ***OR*** $702\text{«km s}^{-1}\text{»}$ ✓

Must see either complete substitution or calculation to at least 3 s.f. for ***MP2***

(c) The photoelectrons are emitted from a sodium surface. Sodium has a work function of 2.3 eV.

Calculate the wavelength of the radiation incident on the sodium. State an appropriate unit for your answer. [3]

Markscheme

$E = 2.3 + 1.4$ ✓

$\lambda = \frac{6.63\times10^{-34}\times3\times10^{8}}{3.7\times1.6\times10^{-19}}$ ✓

$= 3.4 \times 10^{-7}\text{ m}$ ***OR*** 340 nm ✓

Must see an appropriate unit to award ***MP3.***

38. [Maximum mark: 6]

In an experiment to demonstrate the photoelectric effect, monochromatic electromagnetic radiation from source A is incident on the surfaces of metal P and metal Q. Observations of the emission of electrons from P and Q are made.

The experiment is then repeated with two other sources of electromagnetic radiation: B and C. The table gives the results for the experiment and the wavelengths of the radiation sources.

Radiation source	Wavelength / 10^{-7} m	Metal P	Metal Q
A	3.0	electrons emitted	electrons emitted
B	6.0	electrons emitted	no electrons emitted
C	8.0	no electrons emitted	no electrons emitted

(a.i) Outline the cause of the electron emission for radiation A. [1]

Markscheme

photon transfers «all» energy to electron ✓

(a.ii) Outline why electrons are never emitted for radiation C. [1]

Markscheme

photon energy is less than both work functions
OR
photon energy is insufficient «to remove an electron» ✓

Answer must be in terms of photon energy.

(a.iii) Outline why radiation B gives different results. [1]

Markscheme

Identifies P work function lower than Q work function ✓

(b) Explain why there is no effect on the table of results when the intensity of source B is doubled. [1]

Markscheme

changing/doubling intensity «changes/doubles number of photons arriving but» does not change energy of photon ✓

(c) Photons with energy 1.1×10^{-18} J are incident on a third metal surface. The maximum energy of electrons emitted from the surface of the metal is 5.1×10^{-19} J.

Calculate, in eV, the work function of the metal. [2]

Markscheme

$5.1 \times 10^{-19} = 1.1 \times 10^{-18} - \phi$ ✓

work function $=$ «$\frac{(11.0-5.1)\times 10^{-19}}{1.6\times 10^{-19}} =$ » 3.7 «eV» ✓

*Award **[2]** marks for a bald correct answer.*

39. [Maximum mark: 1]

Monochromatic light is incident on a metal surface and electrons are released. The intensity of the incident light is increased. What changes, if any, occur to the rate of emission of electrons and to the kinetic energy of the emitted electrons?

	Rate of emission of electrons	Kinetic energy of the emitted electrons
A.	increase	increase
B.	decrease	no change
C.	decrease	increase
D.	increase	no change

[1]

Markscheme

D

40. [Maximum mark: 1]

A photon has a wavelength λ. What are the energy and momentum of the photon?

	Energy of photon	Momentum of photon
A.	$\frac{hc}{\lambda}$	$\frac{h}{\lambda}$
B.	$\frac{hc}{\lambda}$	$\frac{\lambda}{h}$
C.	$\frac{h\lambda}{c}$	$\frac{h}{\lambda}$
D.	$\frac{h\lambda}{c}$	$\frac{\lambda}{h}$

[1]

Markscheme

A

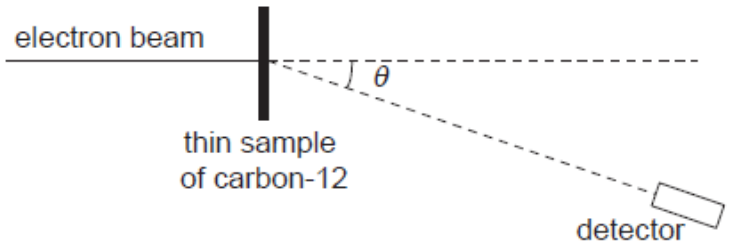
41. [Maximum mark: 9]

The de Broglie wavelength λ of a particle accelerated close to the speed of light is approximately

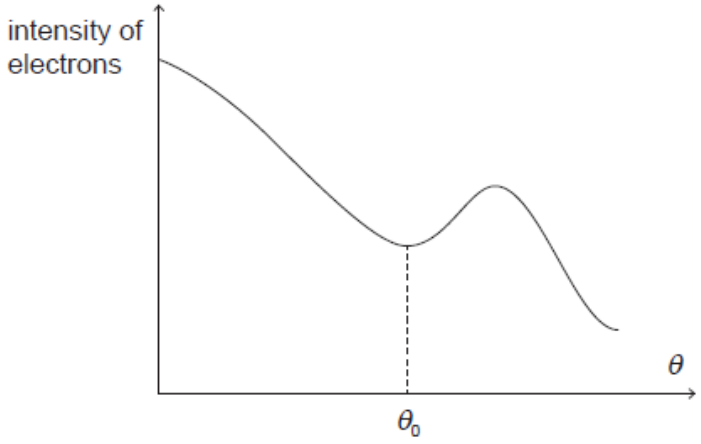
$$\lambda \approx \frac{hc}{E}$$

where E is the energy of the particle.
A beam of electrons of energy $4.2 \times 10^8\ \text{eV}$ is produced in an accelerator.

The electron beam is used to study the nuclear radius of carbon-12. The beam is directed from the left at a thin sample of carbon-12. A detector is placed at an angle θ relative to the direction of the incident beam.



The graph shows the variation of the intensity of electrons with θ. There is a minimum of intensity for $\theta = \theta_0$.



(a) Show that the wavelength of an electron in the beam is about $3 \times 10^{-15}\,\text{m}$. [1]

Markscheme

$\lambda = \frac{6.63\times10^{-34}\times3\times10^{8}}{1.60\times10^{-19}\times4.2\times10^{8}}$ **OR** $= 2.96 \times 10^{-15}$«m» ✓

Answer to at least 2 s.f. (i.e. 3.0)

(b(i)) Discuss how the results of the experiment provide evidence for matter waves. [2]

Markscheme

«the shape of the graph suggests that» electrons undergo diffraction «with carbon nuclei» ✓

only waves diffract ✓

(b(ii)) The accepted value of the diameter of the carbon-12 nucleus is $4.94 \times 10^{-15}\,\text{m}$. Estimate the angle θ_0 at which the minimum of the intensity is formed. [2]

Markscheme

$\sin\theta_0 = \frac{2.96\times10^{-15}}{4.94\times10^{-15}}$« $= 0.599$» ✓

37« degrees » **OR** $0.64/0.65$« rad » ✓

(b(iii)) Outline why electrons with energy of approximately $10^7\ \text{eV}$ would be unsuitable for the investigation of nuclear radii. [2]

Markscheme

the de Broglie wavelength of electrons is «much» longer than the size of a nucleus ✓

hence electrons would not undergo diffraction

OR

no diffraction pattern would be observed ✓

(c) Experiments with many nuclides suggest that the radius of a nucleus is proportional to $A^{\frac{1}{3}}$, where A is the number of nucleons in the nucleus. Show that the density of a nucleus remains approximately the same for all nuclei. [2]

Markscheme

volume of a nucleus proportional to $\left(A^{\frac{1}{3}}\right)^3 = A$ ***AND*** mass proportional to A ✓

the ratio $\frac{\text{mass}}{\text{volume}}$ independent of A «hence density the same for all nuclei» ✓

Both needed for MP1

42. [Maximum mark: 1]
Three observations of the behaviour of electrons are

I. electron emission as a result of the photoelectric effect
II. electron diffraction as an electron interacts with an atom
III. emission of radio waves as a result of electrons oscillating in a conductor.

Which observations are evidence that the electron behaves as a particle?

A. I and II only

B. I and III only

C. II and III only

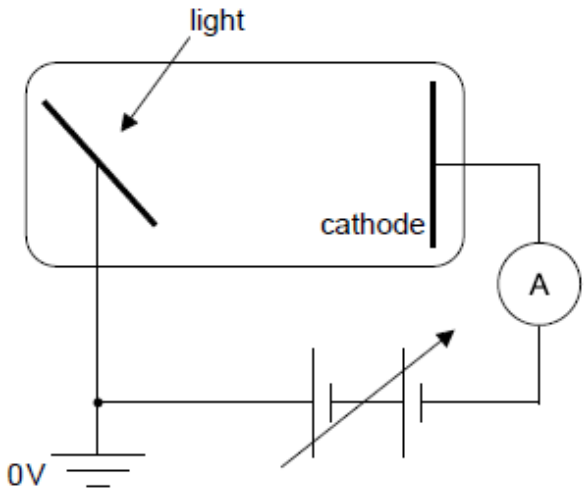
D. I, II and III [1]

Markscheme
B

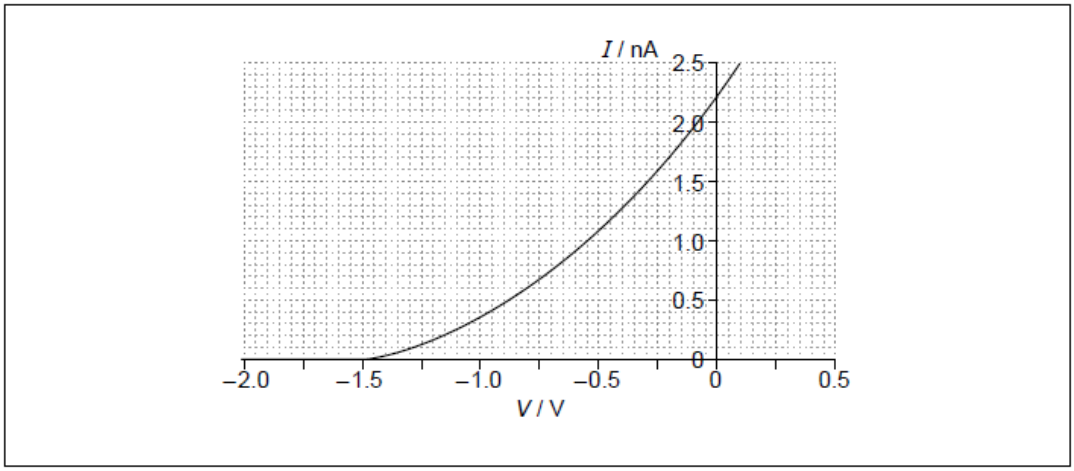
43. [Maximum mark: 9]

Monochromatic light of very low intensity is incident on a metal surface. The light causes the emission of electrons almost instantaneously. Explain how this observation

In an experiment to demonstrate the photoelectric effect, light of wavelength 480 nm is incident on a metal surface.



The graph shows the variation of the current I in the ammeter with the potential V of the cathode.



(a(i)) does not support the wave nature of light. [2]

Markscheme

«low intensity light would» transfer energy to the electron at a low rate/slowly ✔
time would be required for the electron «to absorb the required energy» to escape/be emitted ✔

NOTE: OWTTE

(a(ii)) does support the photon nature of light. [2]

Markscheme

«in the photon theory of light» the electron interacts with a single photon ✔
and absorbs all the energy ***OR*** and can leave the metal immediately ✔

NOTE: Reference to photon-electron collision scores MP1

(b(i)) Calculate, in eV, the work function of the metal surface. [3]

Markscheme

$\phi = \frac{hc}{\lambda} - E_{\mathrm{K}}$ ✔

$E_{\mathrm{K}} = 1.5 \text{« eV »}$ ✔

$\phi = \text{«} \frac{1.24 \times 10^{-6}}{480 \times 10^{-9}} - 1.5 = \text{»} 1.1 \text{« eV »}$ ✔

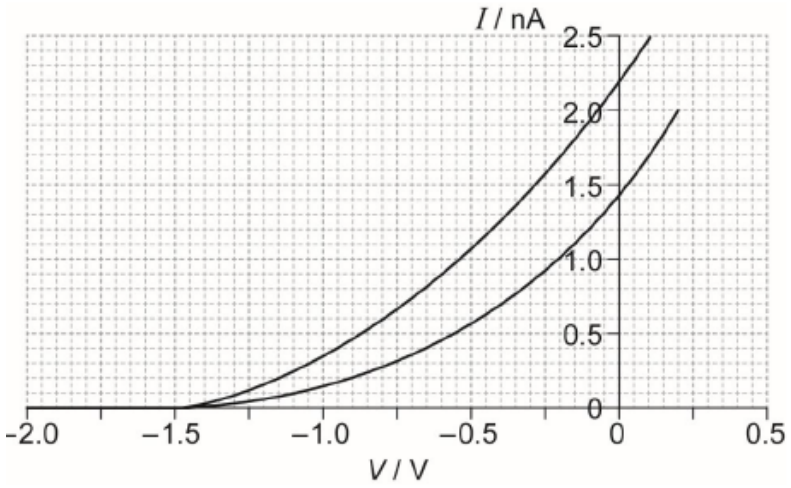
NOTE: Allow reading from the graph of $E_k = 1.4$ leading to an answer of 1.2 «eV».

(b(ii)) The intensity of the light incident on the surface is reduced by half without changing the wavelength. Draw, on the graph, the variation of the current I with potential V after this change. [2]

Markscheme

similar curve lower than original ✔

with same horizontal intercept ✔



44. [Maximum mark: 1]

A metallic surface is first irradiated with infrared radiation and photoelectrons are emitted from the surface. The infrared radiation is replaced by ultraviolet radiation of the same intensity.

What will be the change in the kinetic energy of the photoelectrons and the rate at which they are ejected?

	Kinetic energy of photoelectrons	Rate of ejected photoelectrons
A.	increase	decrease
B.	decrease	decrease
C.	increase	constant
D.	decrease	constant

[1]

Markscheme

A

45. [Maximum mark: 1]

Photons of a certain frequency incident on a metal surface cause the emission of electrons from the surface. The intensity of the light is constant and the frequency of photons is increased. What is the effect, if any, on the number of emitted electrons and the energy of emitted electrons?

	Number of emitted electrons	Energy of emitted electrons
A.	no change	no change
B.	decrease	increase
C.	decrease	no change
D.	no change	increase

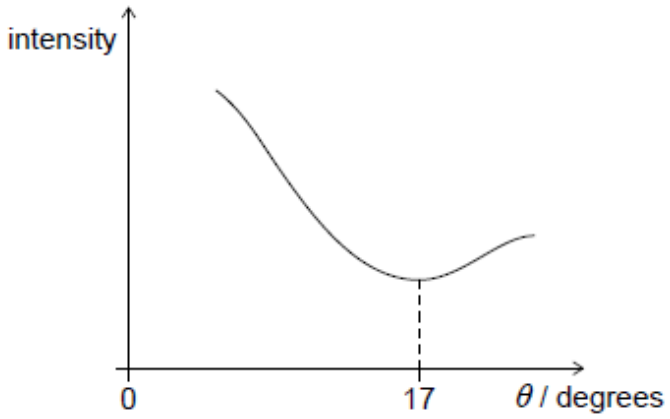
[1]

Markscheme

B

46. [Maximum mark: 10]

In an experiment to determine the radius of a carbon-12 nucleus, a beam of neutrons is scattered by a thin film of carbon-12. The graph shows the variation of intensity of the scattered neutrons with scattering angle. The de Broglie wavelength of the neutrons is 1.6×10^{-15} m.



A pure sample of copper-64 has a mass of 28 mg. The decay constant of copper-64 is 5.5×10^{-2} hour^{-1}.

(a) Suggest why de Broglie's hypothesis is **not** consistent with Bohr's conclusion that the electron's orbit in the hydrogen atom has a well defined radius. [2]

Markscheme

«de Broglie's hypothesis states that the» electron is represented by a wave ✔

therefore it cannot be localized/it is spread out/it does not have a definite position ✔

Award MP1 for any mention of wavelike property of an electron.

(bi) Estimate, using the graph, the radius of a carbon-12 nucleus. [2]

Markscheme

«$d \sin\theta = \lambda \Rightarrow$» $d = \frac{1.6\times10^{-15}}{\sin 17^\circ} / 5.47 \times 10^{-15}$ «m» ✔

$R = \frac{d}{2} \approx 2.7/2.8 \times 10^{-15}$ «m» ✔

(bii) The ratio $\frac{\text{volume of a nucleaus of mass number } A}{\text{volume of a nucleon}}$ is approximately A.

Comment on this observation by reference to the strong nuclear force. [2]

Markscheme

this implies that the nucleons are very tightly packed/that there is very little space in between the nucleons ✔

because the nuclear force is stronger than the electrostatic force ✔

(ci) Estimate, in Bq, the initial activity of the sample. [2]

Markscheme

number of nuclei is $\frac{28\times10^{-3}}{64} \times 6.02 \times 10^{23} / 2.63 \times 10^{20}$ ✔

$A =$«$\lambda N = 2.63 \times 10^{20} \times \frac{5.5\times10^{-2}}{3600}$» 4.0×10^{15} «Bq» ✔

(cii) Calculate, in hours, the time at which the activity of the sample has decreased to one-third of the initial activity. [2]

Markscheme

$\frac{1}{3} = e^{-\lambda t}$ ✔

t = 20«hr» ✔

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